



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Autumn Semester Examination 2023-24

Date of Examination: _____ Session: (FN/AN) _____ Duration: 3 Hrs Full Marks: 60
 Subject No.: MF41601 Subject: Soft Computing
 Department/Center/School: Mechanical Engineering Department
 Specific charts, graph paper, log book etc., required NA
 Special Instructions (if any) : Answer all the questions. Assume suitable data, if necessary.

Marks Distributions Question-wise: 10+14+15+21=60

Q.1. Each question will carry 2 marks. There is no negative marking.

(i) Let us consider two fuzzy sets as follows:

$$A(x) = \{(x_1, 0.4), (x_2, 0.5), (x_3, 0.6)\}$$

$$B(x) = \{(x_1, 0.3), (x_2, 0.4), (x_3, 0.5)\}$$

Their Algebraic sum $A(x) + B(x)$ is determined as

- (a) $\{(x_1, 0.50), (x_2, 0.70), (x_3, 0.90)\}$
- (b) $\{(x_1, 0.06), (x_2, 0.12), (x_3, 0.20)\}$
- ☒ (c) $\{(x_1, 0.58), (x_2, 0.70), (x_3, 0.80)\}$
- (d) $\{(x_1, 0.10), (x_2, 0.20), (x_3, 0.30)\}$

(ii) Let us consider two fuzzy sets as follows:

$$A(x) = \{(x_1, 0.4), (x_2, 0.5), (x_3, 0.6)\}$$

$$B(x) = \{(x_1, 0.5), (x_2, 0.6), (x_3, 0.7)\} \quad \bar{B}(x) = \{(x_1, 0.5), (x_2, 0.4), (x_3, 0.3)\}$$

Their algebraic difference, $A(x) - B(x)$ is determined as

- (a) $\{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3)\}$
- (b) $\{(x_1, 0.5), (x_2, 0.5), (x_3, 0.7)\}$
- (c) $\{(x_1, 0.2), (x_2, 0.2), (x_3, 0.2)\}$
- ☒ (d) $\{(x_1, 0.4), (x_2, 0.4), (x_3, 0.3)\}$

(iii) Let us consider a fuzzy set $A(x)$ as follows: $A(x) = \{(x_1, 0.09), (x_2, 0.16), (x_3, 0.25)\}$. Its DILATION is determined as

- ☒ (a) $\{(x_1, 0.3), (x_2, 0.4), (x_3, 0.5)\}$
- (b) $\{(x_1, 0.9), (x_2, 0.6), (x_3, 0.2)\}$
- © $\{(x_1, 0.1), (x_2, 0.6), (x_3, 0.5)\}$

(d) $\{(x_1, 0.7), (x_2, 0.4), (x_3, 0.1)\}$

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(iv) Which one of the following statements is FALSE?

- (a) Supervised learning of Neural Networks can be implemented in incremental mode.
- (b) Supervised learning of Neural Networks can be implemented in batch mode.
- (c) Supervised learning of Neural Networks can be implemented using the concept of Reward (Success) and Punishment (Failure).
- (d) Batch mode of training can yield an adaptive Neural Network.

(v) In generalized delta rule, learning rate (η) is allowed to vary in the range of

(a) $(-1.0, +1.0)$

☒ (b) $(0.5, +1.5)$

(c) $(0.0, +1.0)$

☒ (d) $(-2.0, +2.0)$

Q. 2. A three-layered feed-forward neural network used to model a two-inputs and one-output process is shown in Fig. 1. The neurons of the input, hidden and output layers are assumed to have the transfer functions represented by $y = a_1 x$, $y = \frac{e^{a_2 x} - e^{-a_2 x}}{e^{a_2 x} + e^{-a_2 x}}$, and $y = x$, respectively. The connecting weights between the input and hidden layers are represented by [V] and those between the hidden and output layers are denoted by [W].

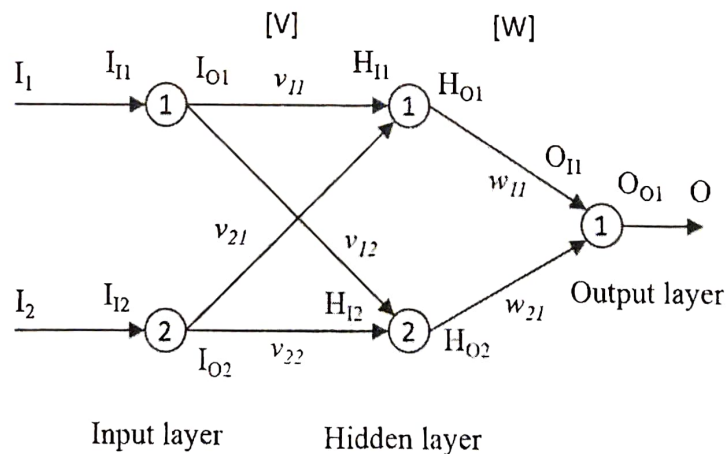


Fig. 1: A three-layered feed-forward neural network.

A binary-coded genetic algorithm (GA) will be used to optimize the parameters of this neural network, in which five bits are utilized to represent each of the real variables. A typical GA-string is shown below.

10010 10100 01011 10101 01110 01010 10011 10100
 v_{11} v_{12} v_{21} v_{22} w_{11} w_{21} a_1 a_2

The connecting weights [V] and [W] vary in the range of $(0.0, 1.0)$ and the parameters: a_1 and a_2 will lie in the range of $(0.5, 1.5)$. Calculate the output for the set of inputs: $I_1=0.6$, $I_2=0.9$. Make comments on how to use this method for evolving an appropriate Neural Network to solve one problem.

Q. 3. To model input-output relationships of a process having two inputs and one output, let us use a fuzzy reasoning tool with Mamdani approach. Let us also assume three linguistic terms, namely Low (L), Medium (M) and High (H) to represent each of two inputs: I_1 , I_2 and one output O . The membership function distributions of the inputs and output are shown in Fig. 2, which are assumed to be symmetrical triangles.

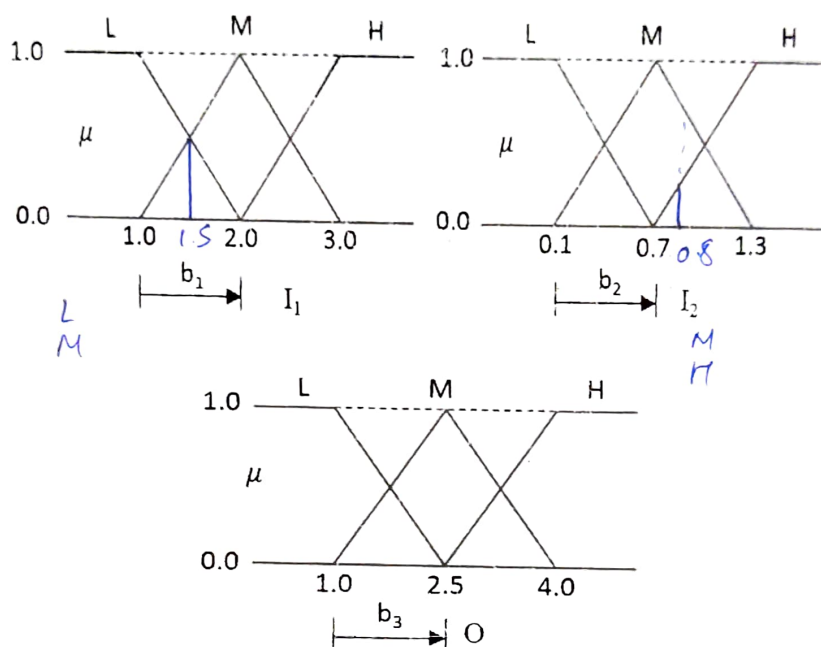


Fig. 2. Membership function distributions of the variables.

A binary-coded genetic algorithm (GA) will be used to evolve the optimized data base (DB) and rule base (RB) of this fuzzy reasoning tool. The base-widths of the triangles are decided by the b values. A typical GA-string is shown below, where five bits are used to represent each of the b values and nine bits are utilized to represent the RB (where 1 and 0 represent the presence and absence of the rules, respectively).

$\underbrace{10111}_{b_1}$ $\underbrace{01101}_{b_2}$ $\underbrace{10100}_{b_3}$ $\underbrace{010110101}_{RB}$

The RB of the fuzzy reasoning tool is shown below.

		I_2		
		L	M	H
I_1	L	L 0	L 1	M 0
	M	L 1	M 1	H 0
	H	H 1	H 0	H 1

$\frac{4}{4}$
Prithvi

Take the ranges of b_1 , b_2 and b_3 as follows:

$$0.5 \leq b_1 \leq 1.5,$$

$$0.4 \leq b_2 \leq 0.9,$$

$$1.0 \leq b_3 \leq 2.0$$

Determine the output for the set of inputs: $I_1 = 1.5$, $I_2 = 0.8$ using the de-fuzzification methods, namely (a) Center of Sums, (b) Mean of Maxima.

Q. 4. Answer all the questions. Each question will carry 3 marks. Answer each question within the words limit of 100.

- (a) Is a fuzzy reasoning tool is nothing but a neural network? Explain.
 - (b) Does a Fuzzy Reasoning Tool have in-built optimization tool? Explain.
 - (c) Which one of the following methods of de-fuzzification can provide the most accurate result: Center of Sums method, Centroid method, Mean of the Maxima method? Explain.
 - (d) Is genetic-neural system basically a Genetic Algorithm? Explain.
 - (e) Neuro-Fuzzy System is basically a fuzzy reasoning tool? Explain.
 - (f) All Expert Systems are Computer Programs, but not all Computer Programs are Expert Systems. Justify the statement.
 - (g) Most of the modern AI-tools work based on the principle of Soft Computing: Justify the statement.
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